```
#'user/rnd
user=> (. rnd nextInt)
1976620372
user=> (. Math PI)
3.141592653589793
user=>
```

Using libraries

Our modest experience with Clojure so far has involved only the core features, or *Clojure Core*, to be technically accurate. You could consider these as the built-in features of Clojure. However, sooner rather than later, you will probably need to access the Clojure features that are packaged as third-party libraries. *Clojure-contrib (http://clojure.github.com/clojure-contrib/)* is a collection of name-spaces, each of which implements various additional features. These libraries are also candidates for inclusion in the Clojure core. There are also a number of other libraries, which can be found at *http://clojure.org/libraries*—and, of course, Github will have a number of others. For a ready-to-use collection of Clojure libraries, see *http://clojars.org*.

In the next example, we will use a Clojure library, Incanter (*https://github.com/liebke/incanter*), which is a Clojure-based R-like statistical-computing environment for the JVM. Let us utilise an easy-to-use build tool for Clojure, called *Leiningen (https://github.com/technomancy/leiningen)*. It is designed to 'not set your hair on fire'—so the Github page claims. It lives up to those words quite well, if my initial experience is any indication.

First, let's set up Leiningen, and then look at a simple use-case of Incanter. Download the *lein* script from *https://github.com/technomancy/leiningen/raw/stable/bin/lein* and *chmod* it to be executable. This shell script will be used to bootstrap Leiningen, and hence the first time you run it, it will download the Leiningen JAR in *$HOME/.lein*. Next, add the location of your *lein* script to your $PATH. Then, create a new project with: *lein new incanter_demo*. This step creates a new sub-directory (*incanter_demo*) for your project, with the following directory structure:

```
.
|-- project.clj
|-- README
|-- src
|   `-- incanter_demo
|       `-- core.clj
`-- test
    `-- incanter_demo
            `-- test
                    `-- core.clj
```

project.clj has the metadata about your project: version information, description, dependencies, etc. As mentioned, we will be using the Incanter Clojure library; hence, we need to add a dependency on Incanter. We also need to set an entry point for our application. Here is what *project.clj* looks like after these modifications:

```
(defproject incanter_demo "1.0.0-SNAPSHOT"
  :description "Demo app using Incanter"
  :dependencies [[org.clojure/clojure "1.2.0"]
                [org.clojure/clojure-contrib "1.2.0"]
        [incanter "1.2.3"]]
:main incanter_demo.core)
```

Next, we will edit the *core.clj* file to the following contents:

```
(ns incanter_demo.core
  (:gen-class))
(use '(incanter core stats charts))
(defn -main []
  (println "Welcome to Incanter demo")
  (view (histogram (sample-normal 1000))))
```

Finally, we will build our Clojure app using *lein uberjar*:

```
Cleaning up.
Downloading: org/clojure/clojure/1.2.0-master-SNAPSHOT/clojure-
1.2.0-master-SNAPSHOT.pom from clojure
Downloading: org/clojure/clojure/1.2.0-master-SNAPSHOT/clojure-
1.2.0-master-SNAPSHOT.pom from clojure-snapshots
Downloading: org/clojure/clojure/1.2.0-master-SNAPSHOT/clojure-
1.2.0-master-SNAPSHOT.pom from clojars
Downloading: org/clojure/clojure/1.2.0-master-SNAPSHOT/clojure-
1.2.0-master-SNAPSHOT.pom from central
Downloading: org/clojure/clojure-contrib/1.2.0-SNAPSHOT/clojure-
contrib-1.2.0-SNAPSHOT.pom from clojure
Downloading: org/clojure/clojure-contrib/1.2.0-SNAPSHOT/clojure-
contrib-1.2.0-SNAPSHOT.pom from clojure-snapshots
Downloading: org/clojure/clojure-contrib/1.2.0-SNAPSHOT/clojure-
contrib-1.2.0-SNAPSHOT.pom from clojars
Downloading: org/clojure/clojure-contrib/1.2.0-SNAPSHOT/clojure-
contrib-1.2.0-SNAPSHOT.pom from central
Copying 38 files to /home/gene/clojure/incanter_demo/lib
Copying 38 files to /home/gene/clojure/incanter_demo/lib
Created /home/gene/clojure/incanter_demo/incanter_demo-1.0.0-
SNAPSHOT.jar
Including incanter_demo-1.0.0-SNAPSHOT.jar
Including jline-0.9.94.jar
Including incanter-charts-1.2.3.jar
Including jlatexmath-0.9.1-20100323.073428-1.jar
Including incanter-1.2.3.jar
Including congomongo-0.1.2-20100502.112537-2.jar
Including jtransforms-0.9.4.jar
Including optimization-0.9.4.jar
Including incanter-excel-1.2.3.jar
Including avalon-framework-4.1.3.jar
Including logkit-1.0.1.jar
Including jcommon-1.0.16.jar
```